



Vignesh K P

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Career Objective

To secure a challenging position in a forward-thinking organization where I can effectively apply my knowledge in VLSI design and IoT to develop smart, reliable, and energy-efficient systems. I aim to contribute to innovative product development by integrating hardware and software solutions. I am eager to continuously learn emerging technologies and enhance my technical and problem-solving abilities to contribute to organizational growth.

Technical Skills

Languages: Python, C, Java, HTML, Verilog HDL

Skills: Web Development.

Frameworks: Arduino

Databases: MySQL

Visualization: Blynk, ThingSpeak, Fritzing

Professional: Team Collaboration, Problem Solving

Experience

Embedded System Programming Intern

May 2026 - Jun 2026

Abhijith Electronic Solutions

- Developed and simulated basic STM32 applications using **STM32CubeMX**, **STM32CubeIDE**, and **Proteus**.
- Configured STM32 microcontroller peripherals and generated initialization code using **STM32CubeMX**.
- Simulated and tested applications including **running LED**, **keypad interfacing**, and other basic GPIO-based circuits.
- Gained hands-on experience with the **STM32 development workflow**, peripheral configuration, simulation, and debugging.

Projects

Design of HEMT For High Frequency Applications

Mar-2026

- Designed and simulated an **AlGaIn/GaN High Electron Mobility Transistor (HEMT)** using **Silvaco ATLAS (TCAD)** for high-frequency RF and microwave applications.
- Analyzed key device characteristics including **I_d-V_d** , **I_d-V_g** , **transconductance (g_m)**, **threshold voltage**, **gate capacitances (C_{gs} , C_{gd})**, and **cutoff frequency (f_T)** through TCAD simulations.
- Utilized **polarization engineering** to achieve high-density **2DEG (Two-Dimensional Electron Gas)** formation, improving electron mobility and current conduction.
- Evaluated the device for **high-frequency communication applications**, demonstrating its suitability for **5G**, **radar**, and **satellite communication systems**.

Design and Comparative Study of 6T and 9T SRAM Cells using 90nm CMOS Technology in Cadence Virtuoso

Jan-2026

- Designed and simulated **6T and 9T SRAM cells** using **90 nm CMOS technology** in Cadence Virtuoso to evaluate memory performance and reliability.
- Performed comparative analysis of **power consumption**, **read/write delay**, **leakage power**, and **read stability** between conventional 6T and proposed 9T SRAM architectures.

- Implemented a **dedicated read port** in the 9T SRAM cell to eliminate read disturbance and improve Static Noise Margin (SNM).
- Validated the designs using **Cadence Virtuoso, DC, and power simulations** under identical operating conditions.

Design of a Low-Power 4-Bit Arithmetic Logic Unit (ALU) Using Cadence Virtuoso (90 nm CMOS) Nov-2025

- Designed and implemented a **low-power 4-bit Arithmetic Logic Unit (ALU)** at the transistor level using **90 nm CMOS technology** in the Cadence Virtuoso platform.
- Developed and integrated CMOS-based **NAND, NOR, XOR, XNOR, Full Adder, Binary-to-Gray Code Converter, 2×1 MUX, and 4×1 MUX** to build a hierarchical ALU architecture.
- Implemented **8 arithmetic and logical operations**: Addition, 1's Complement, XOR, NOR, XNOR, NAND, OR, and Binary-to-Gray Code Conversion using a multiplexer-based control architecture.
- Verified functionality through **Cadence Virtuoso simulations** and evaluated power, propagation delay, and silicon area

Academic Qualification

Sethu Institute of Technology , Virudhunagar B.E. in Electronics and Communication Engineering CGPA: 8.671	2023-Present
Nadar Higher Secondary School , Madurai Higher Secondary Education - 72.5%	2023

Certifications

- Python - Guvi
- Introduction to MS Excel - Coursera
- Introduction to Internet of Things - NPTEL(Elite)
- Cloud Computing - NPTEL
- Programming in C – Codechef

Accepted Publications

- **Performance Analysis of 6T and 9T SRAM Cells in 90 nm CMOS Technology** – Accepted for presentation at the **International Conference on Smart Systems for Electrical, Electronics, Communication and Computer Engineering (IC(SEC)² 2026)**, PSG Institute of Technology and Applied Research, Coimbatore, India, Aug. 28-29, 2026.
- **Design and Optimization of Low Power 4-Bit ALU Using CMOS Technology in Cadence Virtuoso Platform** – Accepted for presentation at the **International Conference on Smart Systems for Electrical, Electronics, Communication and Computer Engineering (IC(SEC)² 2026)**, PSG Institute of Technology and Applied Research, Coimbatore, India, Aug. 28-29, 2026.